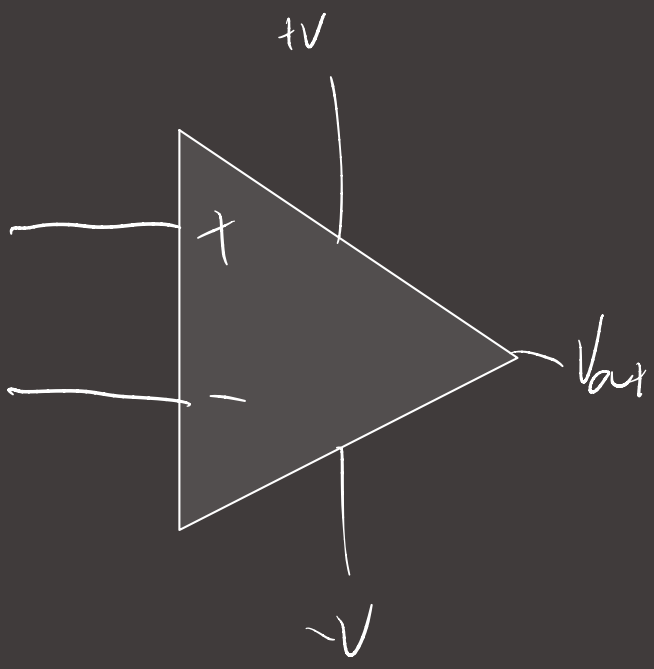
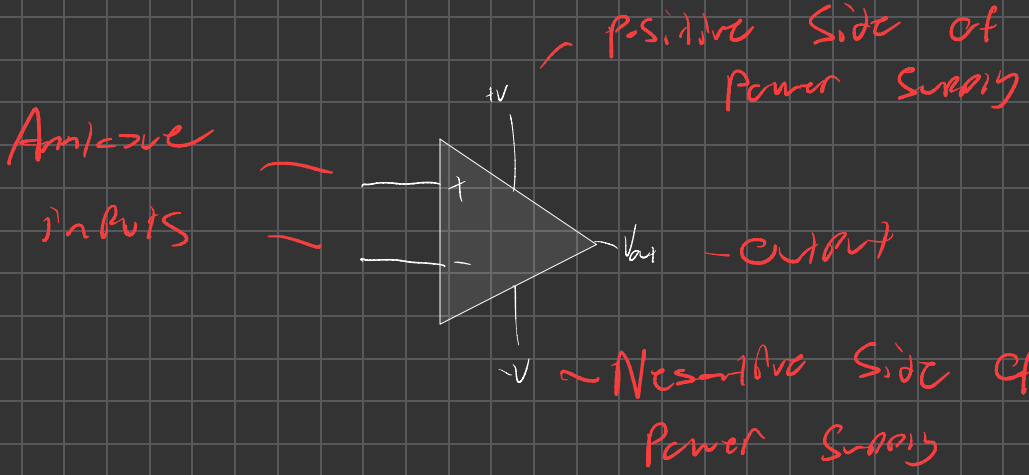
Chance Miller

OP-AMP Comparator

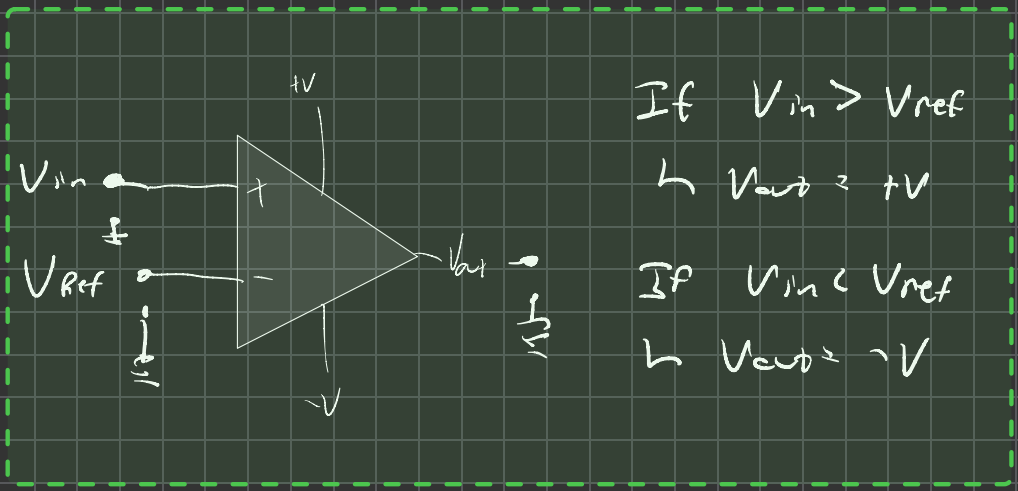


An OP-AMP (Operational Amplifier) is used for many things:

- DC Amplification
- Signal Filtering: 
- Mathematical operations ($\int$, $-$, $\partial/\partial x$, Σ)

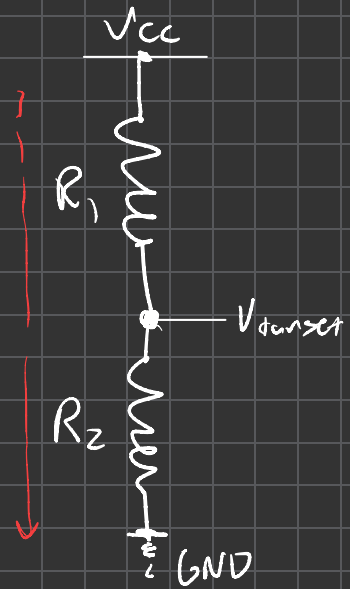


One application of an OP-Amp is a Comparator, which takes two analog input voltages, and returns a 1 or 0 based on their comparison



The output is limited by the OP-Amp's Power Supply, that is, $V_{out} \not> V_+$, or $V_{out} \not< V_-$

We can compare to a fixed voltage by chaining two resistors together



$$R_{total} = R_1 + R_2$$

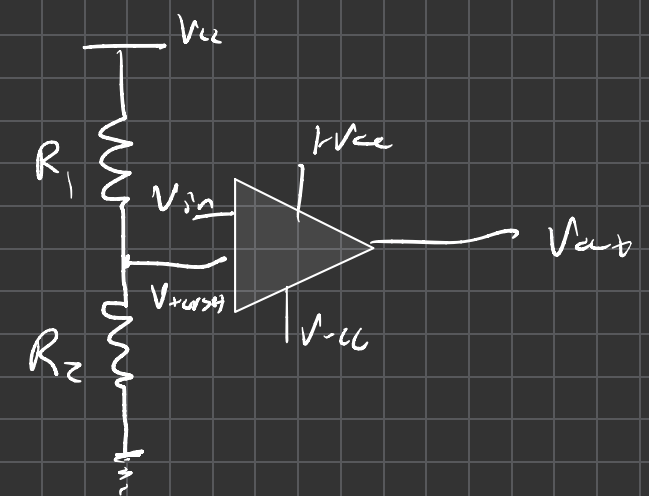
$$\hookrightarrow V_{CC} = I R_{total} \rightarrow I = \frac{V_{CC}}{R_1 + R_2}$$

$$V_{target} = I R_2$$

$$= \frac{V_{CC}}{R_1 + R_2} \cdot R_2$$

$$\boxed{V_{target} = \frac{R_2}{R_1 + R_2} V_{CC}}$$

Chaining this into an OP-AMP...

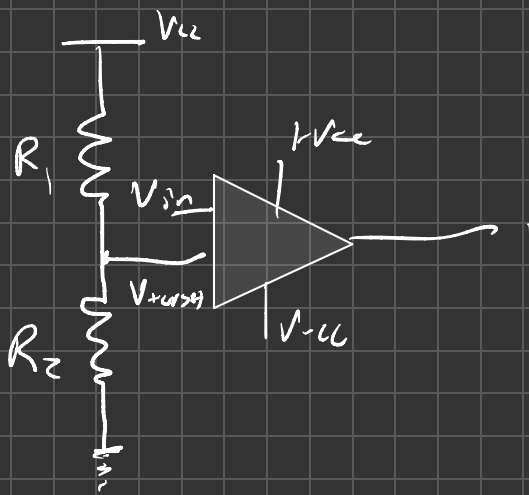


We can compare some input voltage to V_{target} which has voltage:

$$\frac{R_2}{R_2 + R_1} V_{CC}$$

Let's say we want V_{in} to be greater than 3V for the op-amp to read high.

If $V_{cc}=12V$, and $I=100\mu A$, we need to find values for R_1 / R_2 such that



$$V_{out} \frac{R_2}{R_2+R_1} V_{cc} = 3V$$

$$I = \frac{V_{cc}}{R_{total}} = \frac{V_{cc}}{R_1+R_2} \rightarrow R_1+R_2 = \frac{V_{cc}}{I} = \frac{12V}{100\mu A}$$

$$= \frac{12V}{1\times10^{-4}A} = 120{,}000\,\Omega$$

$$\Rightarrow \frac{R_2}{R_1+R_2} V_{cc} = 3V \rightarrow \frac{R_2}{120{,}000\,\Omega} 12V = 3V$$

$$\rightarrow \frac{R_2}{120{,}000} = \frac{1}{4} \rightarrow R_2 = 30{,}000\ \Omega$$

$$R_1 + R_2 = 120{,}000\ \Omega \rightarrow R_1 = 90{,}000\ \Omega$$

So for our circuit to work as intended, $R_1 = 90\ k\Omega$, and $R_2 = 30\ k\Omega$

